

SAFETY DATA SHEET (GHS)

alpha-Methylstyrene (AMS) | CAS: 98-83-9 | Formula: C9H10

PTT Phenol Train II (PPCL), Map Ta Phut, Thailand — HAZOP PSI Document | Issued: 2026-06-13 | SDS Version 1.0

DATA SOURCES CONSULTED

- ECHA Classification & Labelling Inventory — alpha-Methylstyrene, CAS 98-83-9 (<https://echa.europa.eu/>)
- NTP Nomination Background — alpha-Methylstyrene (CASRN 98-83-9), <https://ntp.niehs.nih.gov/>
- Sigma-Aldrich Safety Data Sheet — alpha-Methylstyrene, product 69168, <https://www.sigmaaldrich.com/>
- PubChem CID 7501 — GHS Classification (Rev.9), <https://pubchem.ncbi.nlm.nih.gov/compound/7501>
- Accessed 2026-06-13

SECTION 1 — IDENTIFICATION

Product identifier: alpha-Methylstyrene (AMS)

CAS Number: 98-83-9

Chemical formula: C9H10

Molecular weight: 118.18 g/mol

Synonyms: 2-Phenylpropene; isopropenylbenzene; 1-methyl-1-phenylethylene

Process role: Major by-product of CHP decomposition (formed via dehydration of dimethylbenzyl alcohol/DMBA) in the cumene-to-phenol process; recovered/hydrogenated back to cumene in some configurations. PTT Phenol Train II (PPCL), Map Ta Phut, Thailand

Emergency telephone: [Plant emergency response number — to be inserted by PPCL HSE]

SECTION 2 — HAZARD IDENTIFICATION

GHS classification: Flam. Liq. 3 (H226); Eye Irrit. 2 (H319); STOT SE 3 (H335); Aquatic Chronic 2 (H411)

Signal word: Warning

Hazard statements (H-codes): H226 (Flammable liquid and vapour); H319 (Causes serious eye irritation); H335 (May cause respiratory irritation); H411 (Toxic to aquatic life with long lasting effects)

Precautionary statements (P-codes): P210, P233, P240, P241, P242, P243, P261, P264, P271, P273, P280, P303+P361+P353, P304+P340, P305+P351+P338, P337+P313, P370+P378, P403+P235, P501

GHS pictograms: GHS02 (Flame), GHS07 (Exclamation mark), GHS09 (Environment)

Other hazards: DATA-QUALITY FLAG: one supplier (Sigma-Aldrich) SDS listing returned in source searches also lists Asp. Tox. 1 (H304), Repr. 2 (H361), and Skin Sens. 1B for this CAS number, in addition to the ECHA-aligned classification used above (H226/H319/H335/H411). These additional categories were NOT included in this SDS's headline classification because they could not be corroborated against the ECHA C&L Inventory harmonised/aggregated entries reviewed for this document — this mirrors a similar data-aggregation discrepancy observed for cumene (CAS 98-82-8, see that SDS Section 2/16). RECOMMEND: manually verify AMS classification (particularly possible aspiration hazard H304, given AMS is a low-viscosity aromatic liquid like cumene, and possible reproductive toxicity/skin sensitisation flags) against the current ECHA C&L Inventory and the specific supplier SDS used on-site before finalising PSI documentation. AMS can also form peroxides on prolonged storage/exposure to air and light, similar to styrene — store with inhibitor and monitor peroxide content for long-term storage.

SECTION 3 — COMPOSITION / INFORMATION ON INGREDIENTS

Substance: alpha-Methylstyrene, CAS 98-83-9, typically >99% purity; may contain polymerisation inhibitor (e.g., TBC — see separate Diamine/TBC SDS) for stabilisation during storage

SECTION 4 — FIRST AID MEASURES

Inhalation: Remove to fresh air. If breathing is difficult, give oxygen. Obtain medical attention if respiratory irritation (H335) persists.

Skin contact: Remove contaminated clothing and wash skin with plenty of water and soap.

Eye contact: Flush immediately with copious water for at least 15 minutes, holding eyelids open. Obtain medical attention (H319).

Ingestion: Rinse mouth. Do not induce vomiting. Obtain medical attention immediately.

Notes to physician: Treat symptomatically; monitor for respiratory tract irritation following significant inhalation exposure

SECTION 5 — FIREFIGHTING MEASURES

Suitable extinguishing media: Dry chemical, CO₂, alcohol-resistant foam, water spray (for cooling)

Unsuitable extinguishing media: High-volume water jet (may spread fire)

Specific hazards: Flammable liquid (H226), flash point ~45-54°C — confirm exact value against the in-use supplier SDS. Vapours may form explosive mixtures with air. Can polymerise exothermically if uninhibited and exposed to heat — runaway polymerisation is a potential fire/explosion escalation hazard for bulk storage.

Special protective equipment for firefighters: Self-contained breathing apparatus (SCBA) and full protective clothing

Hazardous combustion products: Carbon monoxide, carbon dioxide, smoke

SECTION 6 — ACCIDENTAL RELEASE MEASURES

Personal precautions: Eliminate ignition sources; ensure adequate ventilation; wear appropriate PPE (Section 8)

Environmental precautions: Prevent entry to drains, watercourses, and soil (H411 — toxic to aquatic life)

Containment and cleanup: Contain spill with inert absorbent material; collect into closed, labelled containers for disposal; avoid generating vapour

Reference to other sections: See Section 7 (Handling/Storage — inhibitor requirements) and Section 10 (polymerisation hazard)

SECTION 7 — HANDLING AND STORAGE

Handling precautions: Use only in well-ventilated areas; avoid ignition sources (electrical equipment, open flames, static discharge — bond/ground transfer equipment); avoid eye contact and inhalation of vapours

Storage conditions: Store in tightly closed containers in a cool, well-ventilated area away from heat, sparks, and open flame; protect from direct sunlight

Temperature limits: Store below the temperature at which polymerisation inhibitor depletion becomes a concern (confirm inhibitor depletion rate vs. storage temperature with supplier); avoid prolonged storage at elevated temperature

Pressure: Atmospheric storage with inert gas blanketing (e.g., nitrogen, CAS 7727-37-9) recommended to reduce oxidation/peroxide formation and exclude moisture

Segregation requirements: Segregate from oxidisers, strong acids, and polymerisation initiators; segregate from heat sources

Special requirements: Verify polymerisation inhibitor (e.g., TBC) concentration is maintained within the supplier's recommended range for the storage duration and temperature — inhibitor depletion in the presence of oxygen-depleted (fully inert) atmospheres can behave differently than in air-saturated systems; coordinate inhibitor management with the Diamine/TBC SDS

Specific end-use recommendations: Periodically test for peroxide content if AMS is stored for extended periods, particularly in the presence of light/air

SECTION 8 — EXPOSURE CONTROLS / PERSONAL PROTECTION

Occupational Exposure Limits (OEL): ACGIH TLV-TWA: 50 ppm (242 mg/m³) for alpha-methylstyrene; confirm Thailand OEL alignment

TWA: 50 ppm (ACGIH)

STEL: 100 ppm (ACGIH, as a ceiling-type guidance for some jurisdictions — confirm against current ACGIH TLV booklet)

Ceiling: See STEL note above — confirm current ACGIH guidance

Skin notation: Not assigned by ACGIH for AMS (confirm against current TLV booklet given the Sigma SDS Skin Sens. 1B flag noted in Section 2)

IARC/NTP/OSHA carcinogen status: NTP has reviewed alpha-methylstyrene for possible nomination; not currently listed as a confirmed human carcinogen by IARC/NTP/OSHA — confirm current status

Engineering controls: Local exhaust ventilation at transfer points and sample points; closed sampling systems preferred

PPE — Respiratory: Organic vapour cartridge respirator for tasks above the OEL; SCBA for emergency response

PPE — Gloves: Chemical-resistant gloves (e.g., nitrile, Viton) — verify breakthrough time against AMS

PPE — Eyes: Chemical splash goggles (H319)

PPE — Body: Flame-resistant work clothing; chemical-resistant apron for transfer operations

Biological monitoring: Not routinely established

SECTION 9 — PHYSICAL AND CHEMICAL PROPERTIES

Appearance: Colourless to slightly yellow liquid

Odour: Sharp, characteristic aromatic odour

Odour threshold: Not established

pH: Not applicable

Melting/Freezing point: -23.2°C

Boiling point (at 1 atm): 165.4°C

Flash point: 45-54°C (closed cup) — value varies by source; confirm against supplier SDS for the specific product used

Evaporation rate: Slower than water

Flammability (solid/gas): Not applicable (liquid)

Flammability limits in air: Lower ~0.9% v/v; Upper ~6.1% v/v (estimated, confirm against authoritative source)

Vapour pressure (at 20°C): ~0.2 kPa (~1.5 mmHg)

Vapour density (air = 1): 4.08

Relative density (water = 1): 0.909 at 20°C

Solubility (water): Practically insoluble (~80 mg/L)

Partition coefficient (log K_{ow}): ~3.1

Auto-ignition temperature: Not well established — confirm; aromatic vinyl monomers of this type typically have autoignition temperatures in the 480-570°C range

Decomposition temperature: Not established; thermal polymerisation can occur at elevated temperature in the absence of adequate inhibitor

Viscosity: ~0.7 mPa·s at 25°C

Explosive properties: Not classified as explosive; vapour/air mixtures within flammable range can ignite

Oxidising properties: Not an oxidiser

Molecular weight: 118.18 g/mol

SECTION 10 — STABILITY AND REACTIVITY

Reactivity: Can undergo exothermic polymerisation, especially if uninhibited and exposed to heat, light, peroxides, or polymerisation initiators

Chemical stability: Stable when adequately inhibited and stored as recommended; uninhibited material or material with depleted inhibitor is at risk of spontaneous polymerisation

Possibility of hazardous reactions: Hazardous polymerisation; can form peroxides on prolonged exposure to air/light (similar to styrene)

Conditions to avoid: Heat, direct sunlight, contact with peroxides/oxidisers, depletion of polymerisation inhibitor

Incompatible materials: Strong oxidisers, polymerisation initiators (e.g., peroxides — relevant given proximity to CHP, CAS 80-15-9, in this process), strong acids

Hazardous decomposition products: Carbon monoxide, carbon dioxide on combustion; polymer/oligomer products if uncontrolled polymerisation occurs

SECTION 11 — TOXICOLOGICAL INFORMATION

Routes of exposure: Inhalation, skin contact, eye contact, ingestion

Acute toxicity — Oral LD₅₀: ~4,900 mg/kg, rat (low acute oral toxicity)

Acute toxicity — Dermal LD₅₀: >5,000 mg/kg, rabbit (low acute dermal toxicity)

Acute toxicity — Inhalation LC50: Not well established for AMS specifically

Skin corrosion/irritation: May cause mild skin irritation; not classified as a corrosive under the ECHA-aligned classification used here (see Section 2 flag re: possible skin sensitisation)

Serious eye damage/irritation: Causes serious eye irritation (H319)

Respiratory/skin sensitisation: Not classified under ECHA-aligned classification used here; one supplier source flags possible Skin Sens. 1B — see Section 2 data-quality flag

Germ cell mutagenicity: Not classified

Carcinogenicity: Not currently classified — NTP nomination background document exists but AMS is not listed as a confirmed carcinogen by IARC/NTP/OSHA

Reproductive toxicity: Not classified under ECHA-aligned classification used here; one supplier source flags possible Repr. 2 — see Section 2 data-quality flag, recommend verification

STOT (single exposure): May cause respiratory irritation (H335)

STOT (repeated exposure): Not classified under primary classification used; confirm against supplier SDS given flags above

Aspiration hazard: Not classified under ECHA-aligned classification used here; one supplier source flags possible Asp. Tox. 1 — see Section 2 data-quality flag, recommend verification given AMS's low viscosity/kinematic properties similar to cumene

SECTION 12 — ECOLOGICAL INFORMATION

Aquatic toxicity (LC50/EC50): Toxic to aquatic life with long-lasting effects (H411) — fish LC50 in the low single-digit mg/L range reported for related aromatic vinyl monomers

Persistence and degradability: Expected to biodegrade under aerobic conditions but not readily; not expected to be persistent

Bioaccumulative potential: Moderate bioaccumulation potential indicated by log Kow ~3.1

Mobility in soil: Moderate mobility; low water solubility limits leaching

Other adverse effects: None specifically identified beyond aquatic toxicity

SECTION 13 — DISPOSAL CONSIDERATIONS

Disposal methods: Dispose of via licensed hazardous waste incineration or treatment facility in accordance with local regulations; do not discharge to drains

Contaminated packaging: Dispose of as hazardous waste or, if uncontaminated/empty, per local regulations after appropriate purging

Regulatory reference: Thailand Hazardous Substance Act B.E. 2535 (verify applicability); local industrial waste regulations

SECTION 14 — TRANSPORT INFORMATION

UN Number: UN2303

Proper shipping name: ISOPROPENYLBENZENE (alpha-METHYLSTYRENE)

Transport hazard class(es): Class 3 (Flammable liquid)

Packing group: III

Environmental hazards (marine pollutant): Yes — marine pollutant (consistent with H411 classification)

Special precautions for user: Ensure adequate ventilation during loading/unloading; verify inhibitor stabilisation prior to transport

Bulk transport (MARPOL/IBC): Subject to IBC Code provisions for bulk marine transport — confirm pollution category

SECTION 15 — REGULATORY INFORMATION

Safety and health regulations: See country-specific notes below

Thailand: Hazardous Substance Act B.E. 2535 — classification depends on concentration/use; verify listing (List 1-4) with Department of Industrial Works.

EU: Registered under REACH (Regulation (EC) No 1907/2006); classified under CLP Regulation (EC) No 1272/2008.

US: Subject to OSHA Hazard Communication Standard 29 CFR 1910.1200 (HazCom 2012, aligned with GHS Rev.7).

ASEAN: Classification consistent with ASEAN GHS implementation (harmonised with UN GHS).

Chemical safety assessment: Yes — performed by upstream registrants under REACH; site-specific CSA recommended as part of PPCL PSI documentation.

SECTION 16 — OTHER INFORMATION

Revision date: 2026-06-13

Revision number: 1.0

Key changes from previous version: Initial issue — compiled for HAZOP PSI (CAS 98-83-9). DATA-QUALITY FLAG: a supplier SDS source returned additional hazard categories (Asp. Tox. 1 H304, Repr. 2 H361, Skin Sens. 1B) beyond the ECHA-aligned classification used here (H226/H319/H335/H411) — flagged for manual verification, analogous to the cumene (CAS 98-82-8) Carc. 1B discrepancy noted in that SDS.

Data sources used in this SDS: See 'Data Sources Consulted' block at top of document

Disclaimer: This SDS was compiled for HAZOP Process Safety Information purposes at PTT Phenol Train II (PPCL), Map Ta Phut, Thailand. Data sourced from published authoritative references (ECHA, PubChem/GHS Rev.9, NIOSH, NTP, supplier SDSs, CAMEO Chemicals, peer-reviewed literature). Verify against current manufacturer SDS before use for regulatory compliance or labelling purposes.

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